

NO SPOILER PLEASE! A TINYBERT MODEL TO DETECT SPOILER REVIEWS ON

NLP FINAL PROJECT FOR THE 2025/26 ACADEMIC YEAR

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P6: THE APPRENTICE MODEL

THE PROBLEM WITH IMDB SPOILER WARNINGS

- Reviews flagged by users as spoilers
- A user can easily mark a review as NO SPOILERS and post them instead.
- Possible solution: browser extension powered by an LLM like BERT.
- **Problem:** Such a model can be very expensive in terms of both memory and inference time, but a smaller model is not as efficient.

★ 9/10

So good and realistic >

SPOILER

This show is so captivating with its characters, their personal dynamics and the reality of stress that comes with working in an ER. I found myself liking all of the characters and found them intriguing enough to bring me back for more until I ran out of episodes.

The level of detail is what impresses me the most. I have no medical background but as the episodes (or hours of the shift) go, you see people becoming increasingly tense with one another and the team dynamic begins to breakdown. I can only assume this was done with the intention of showing how stressful and draining a 12 hour ER shift can be.

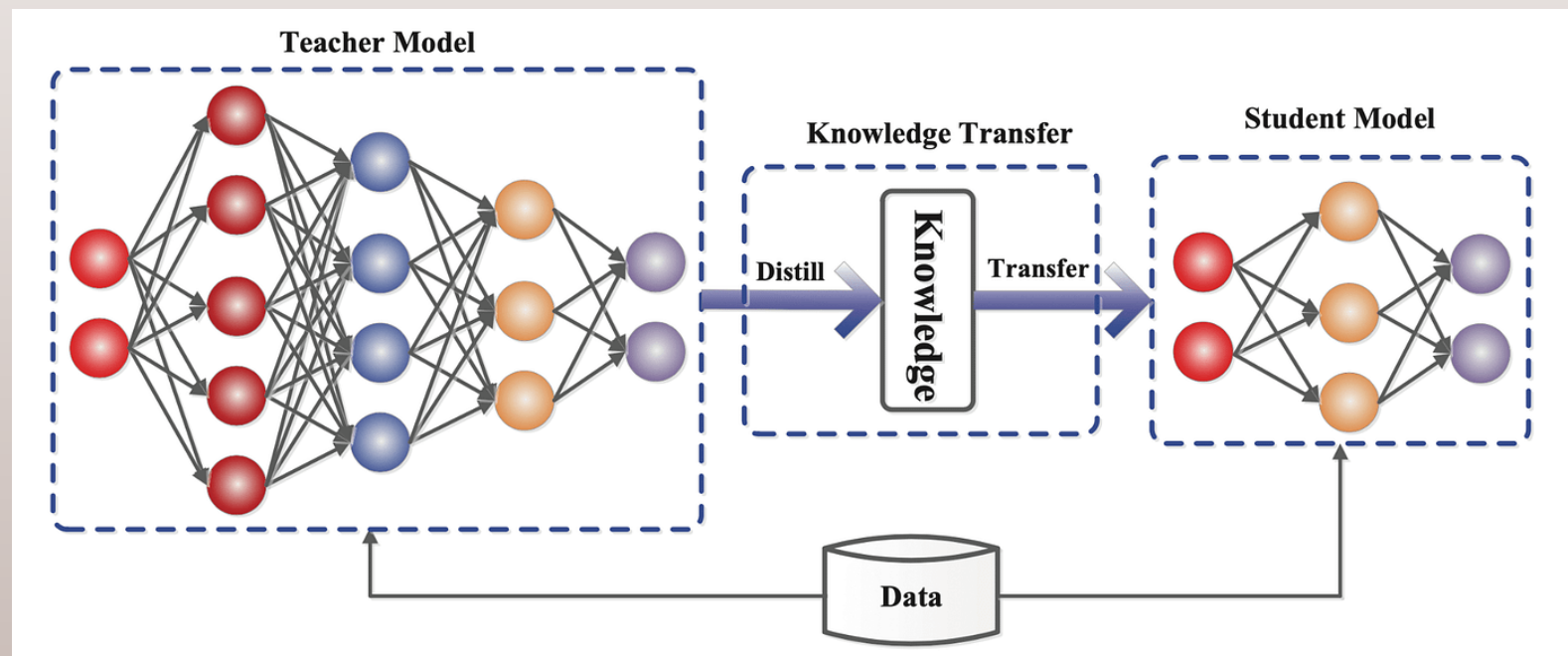
I need a second season, now!

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INTRODUCTION TO KNOWLEDGE DISTILLATION

- Main idea: transfer the knowledge of a big LLM model (teacher) to a smaller LLM model (student).
- Objective: obtain a lighter student model but with performance very close to that of the teacher model.
- How: train the student model to imitate the prediction layer, attention layers and hidden states of the fine-tuned teacher model.

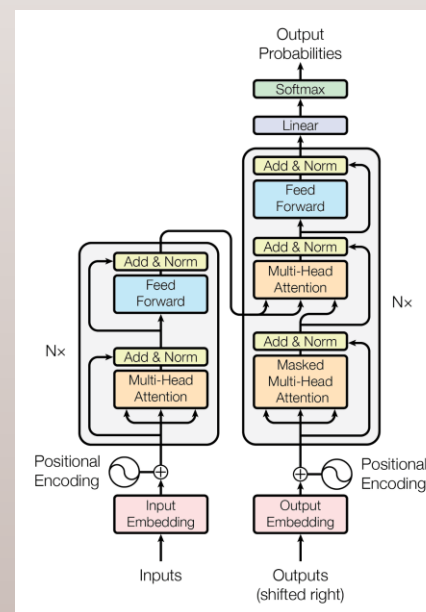


RESEARCH QUESTION AND METHODOLOGY

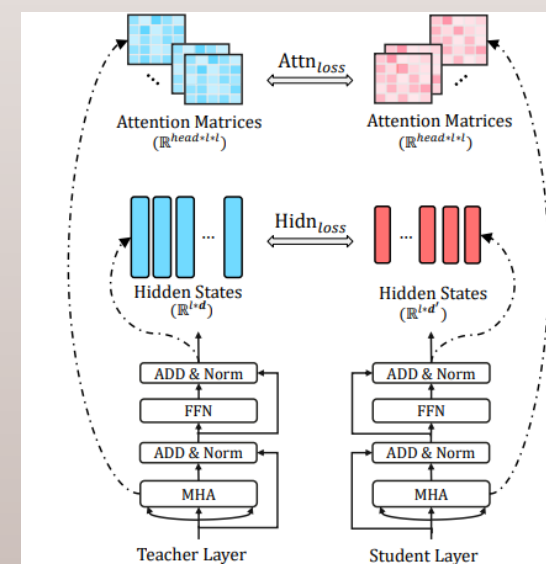
To what extent can a highly compressed language model (TinyBERT) maintain not only the predictive performance of a larger teacher model (BERT-base) but also its internal reasoning and contextualization capabilities in a highly ambiguous task like spoiler detection?

IMDb

1. Data collection and preprocessing



2. Teacher Fine Tuning



3. Two-phase student distillation

DATASET & PREPROCESSING

- Dataset: IMDB Spoiler Dataset on Kaggle
 - Over 500000 labeled reviews
 - A random sample of 300000 reviews has been used
 - Split: 80% train, 10% validation and 10% test
 - Split by movie_ID to prevent leakages
- Two main problems:
 - 1. Imbalance on labels (73% No Spoilers vs 27% Spoilers)
 - Solution: Weighted Cross Entropy Loss
 - 2. Length of the reviews can exceed BERT's 512-token limit
 - Solution: Head+Tail truncation strategy.

Algorithm 1 Head+Tail Truncation Strategy for BERT

Input: Review text R

Output: Token sequence T_{final} (max 512 tokens)

// Tokenize text using the BERT vocabulary

1 $T \leftarrow \text{Tokenize}(R)$

// Add special tokens

2 $T_{spec} \leftarrow [\text{CLS}] \oplus T \oplus [\text{SEP}]$ $N \leftarrow \text{Length}(T_{spec})$

3 **if** $N > 512$ **then**

 // Retain the first 128 tokens after [CLS]

4 $T_{head} \leftarrow$ first 128 tokens of T // Retain the last 382 tokens before [SEP]

5 $T_{tail} \leftarrow$ last 382 tokens of T // Concatenate for exactly 512 tokens

6 $T_{final} \leftarrow [\text{CLS}] \oplus T_{head} \oplus T_{tail} \oplus [\text{SEP}]$

7 **else**

 // Keep natural length, to be dynamically padded at batch time

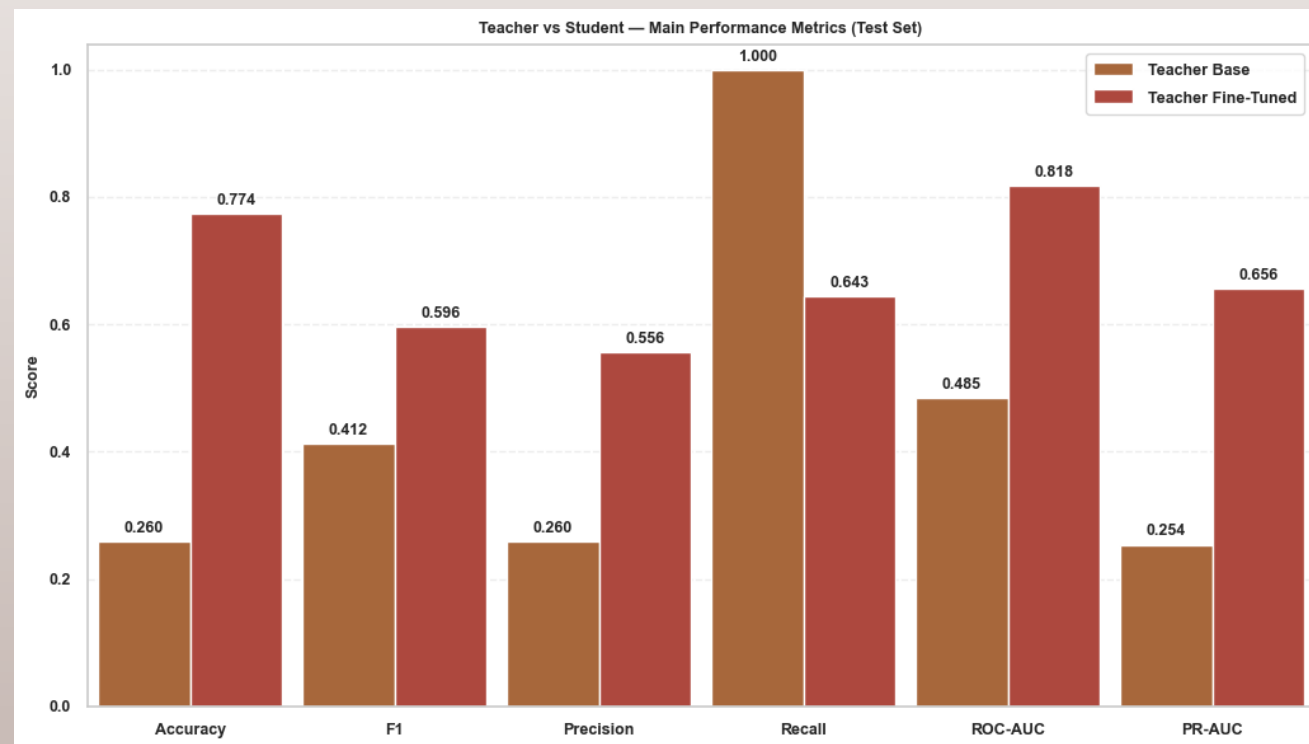
8 $T_{final} \leftarrow T_{spec}$

9 **end**

10 **return** T_{final}

TEACHER FINE TUNING

- Model: BERT-base-uncased (110M Parameters)
- Training protocol:
 - Epochs: 3
 - Batch size: 24
 - Warmup ratio: 0.03
 - Metric for best model: PR-AUC on val set
 - Loss: Weighted Cross-Entropy to penalize spoiler misclassification (calculated on train set)
 - Optimizer: AdamW with Layer-wise Learning Rate Decay to avoid catastrophic forgetting
 - LR base = $2e-5$
 - Decay factor: 0.95



STUDENT DISTILLATION

- Model: TinyBERT_General_4L_312D (14.5M Parameters)
- Phase 1: Intermediate Layers Distillation
 - Aligns embeddings, hidden states and attention matrices between teacher and student
 - Uses learnable projections to handle dimension mismatch
- Phase 2: Prediction Layer Distillation
 - Combines softened teacher logits with weighted hard labels
 - Adaptive coefficient: teaches the student to trust the teacher more when its predictions are correct

$$L_{TD1}(x) = \lambda_0 L_{embd}(x) + \sum_{m=1}^M \lambda_m (L_{hidn}^{(m)}(x) + L_{attn}^{(m)}(x))$$

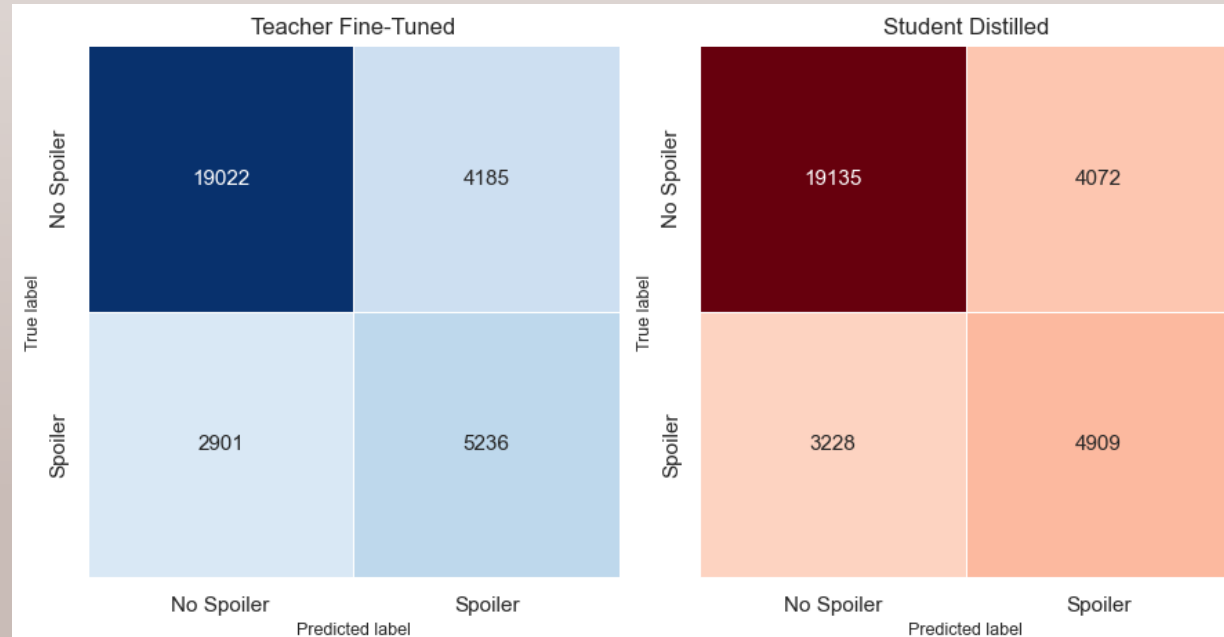
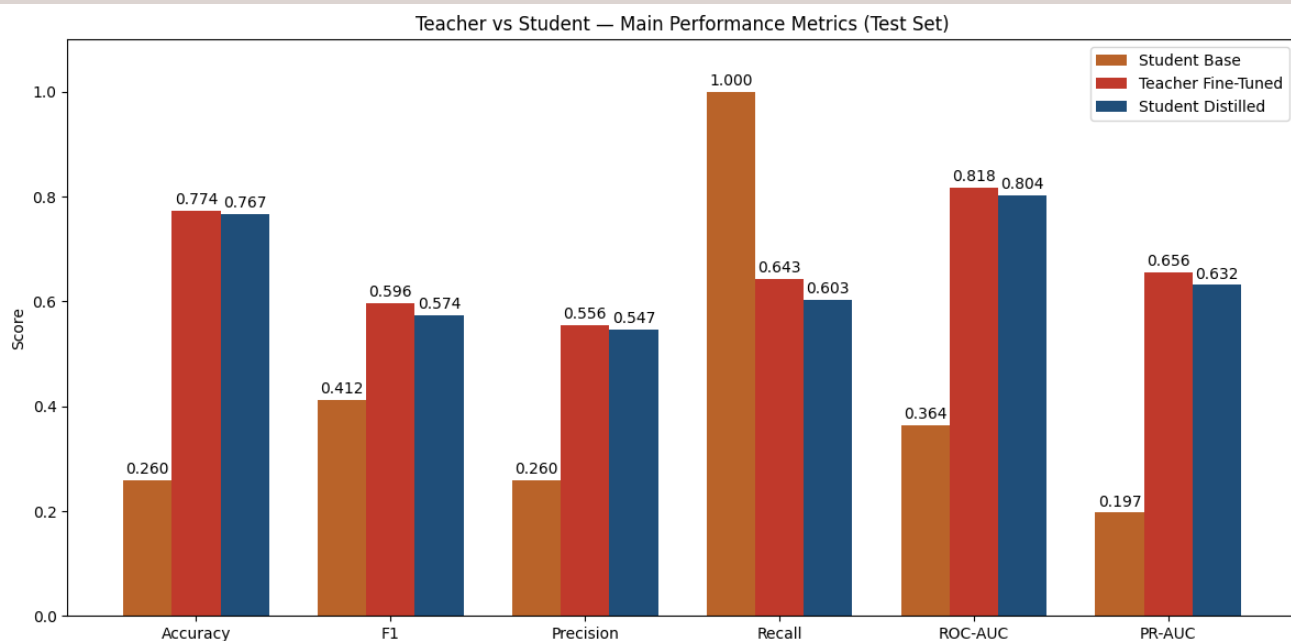
$$L_{TD-2} = \rho(x) \cdot \underbrace{CE(z_T/T, z_S/T)}_{\text{Soft Loss}} \cdot T^2 + (1 - \rho(x)) \cdot \underbrace{WCE(z_S, y_{true})}_{\text{Weighted Hard Loss}}$$

Model	Layers	Attention heads	Embeddings	Hidden states	Total params
BERT-base	12	12	768	768	110M
TinyBERT	4	12	312	312	14.5M

Phase	Epochs	Batch size	Learning rate	Metric for best model
1	8	32 (8 x 4)	5e - 5	-
2	3	32 (16 x 2)	2e - 5	PR-AUC

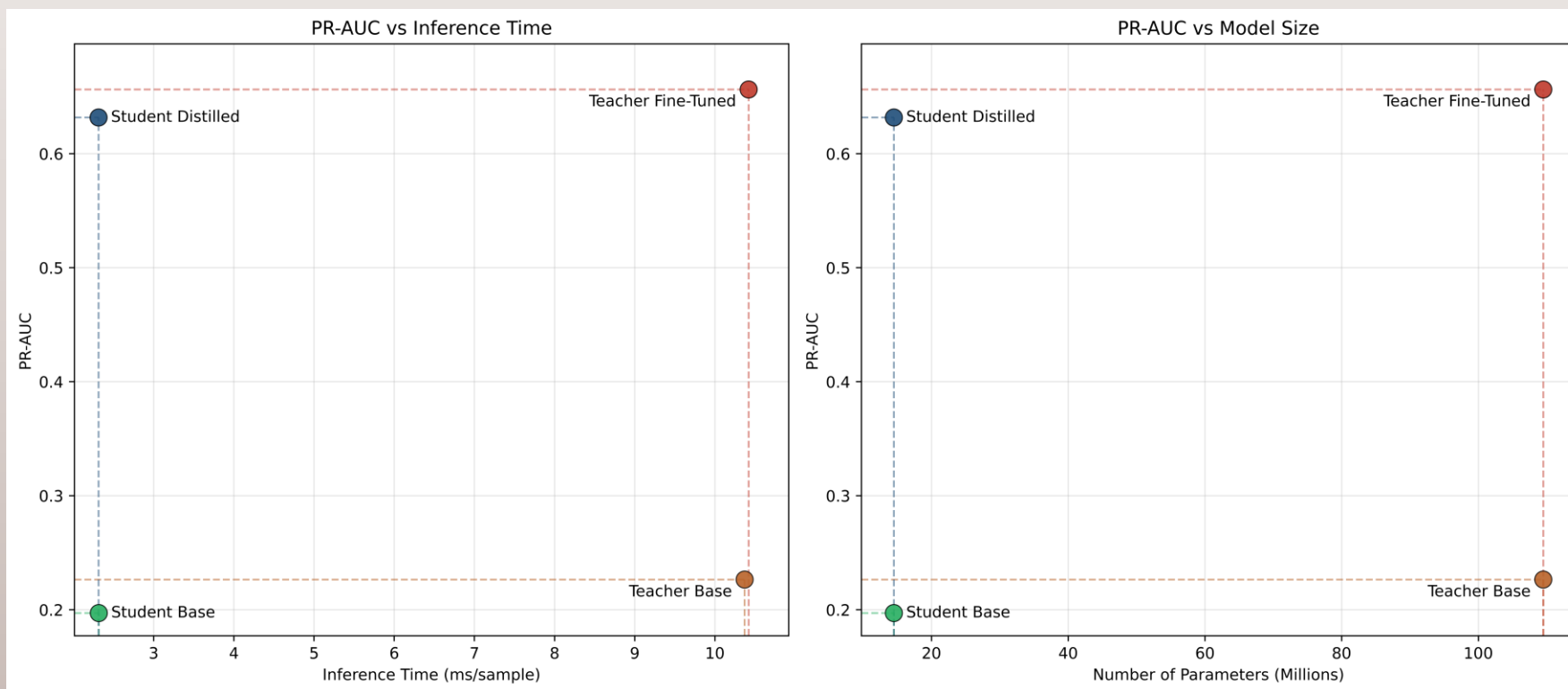
RESULTS: PERFORMANCE DROP

- Distillation process introduces a very limited performance drop
- Metrics (teacher vs distilled student):
 - Only a 3.74% decrease in PR-AUC
 - Slight increase in FN (small drop in recall), but overall highly consistent confusion matrices
- Baseline: vastly outperforms the COTS TinyBERT baseline, which suffered from high FP



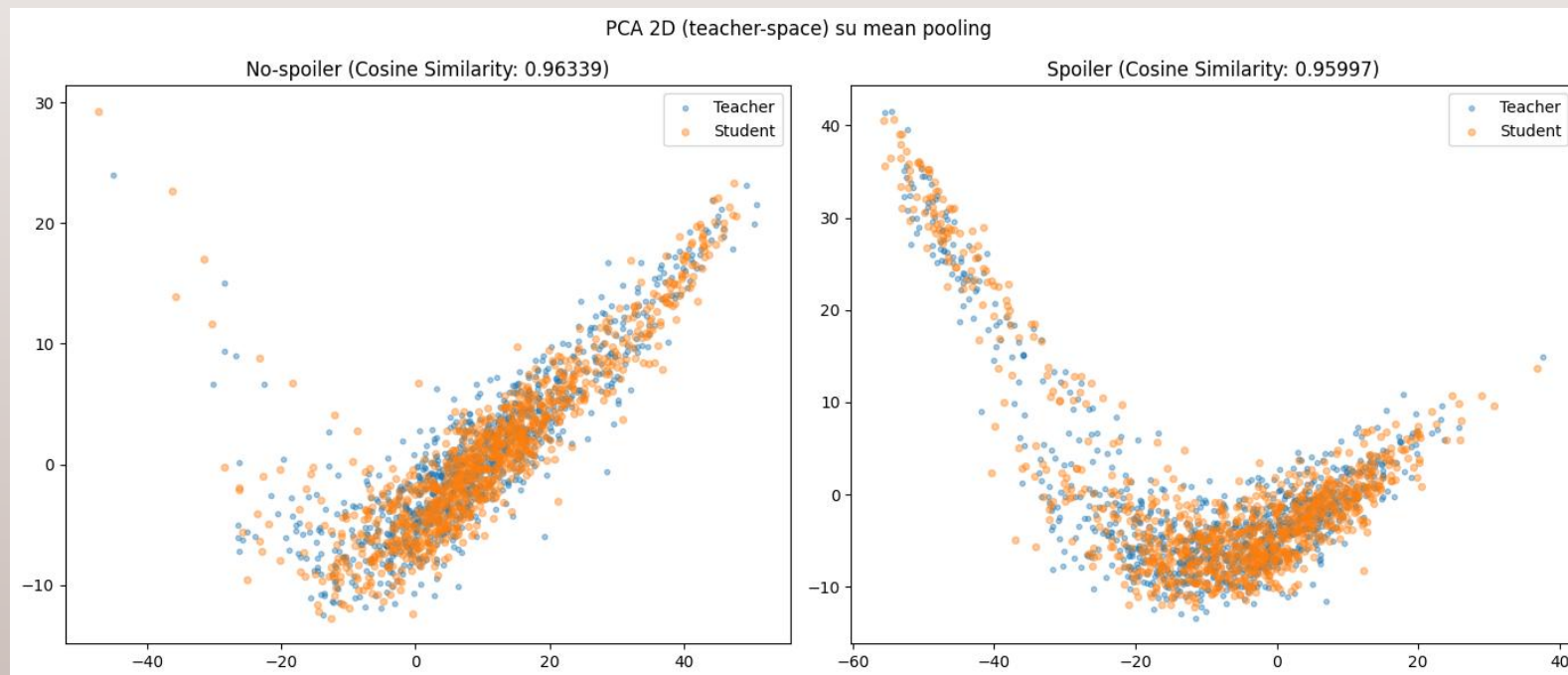
RESULTS: COMPUTATIONAL SAVINGS

- Efficiency gains: the distilled student achieves a highly favorable efficiency-performance trade-off.
- Parameter reduction: 86.76% fewer parameters.
- Speedup: 77.81% reduction in inference time per sample compared to the fine-tuned teacher.



RESULTS: ALIGNMENT & INTERPRETABILITY (I)

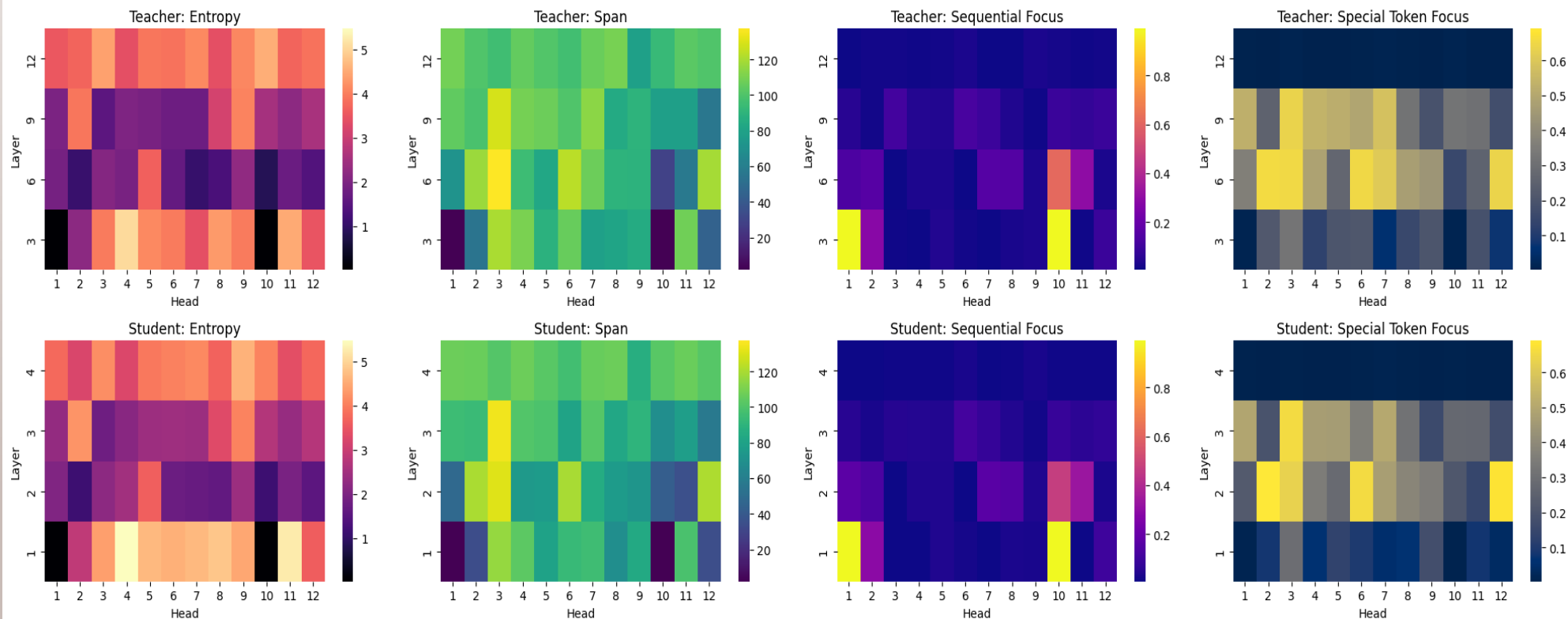
- Last-layer student states are projected into the teacher space.
- Mean pooling and a shared 2D PCA are computed on a balanced test subset.
- Teacher and student distributions largely overlap for both classes.
- Average cosine similarity is around 0.96, suggesting similar representation geometry.



RESULTS: ALIGNMENT & INTERPRETABILITY (2)

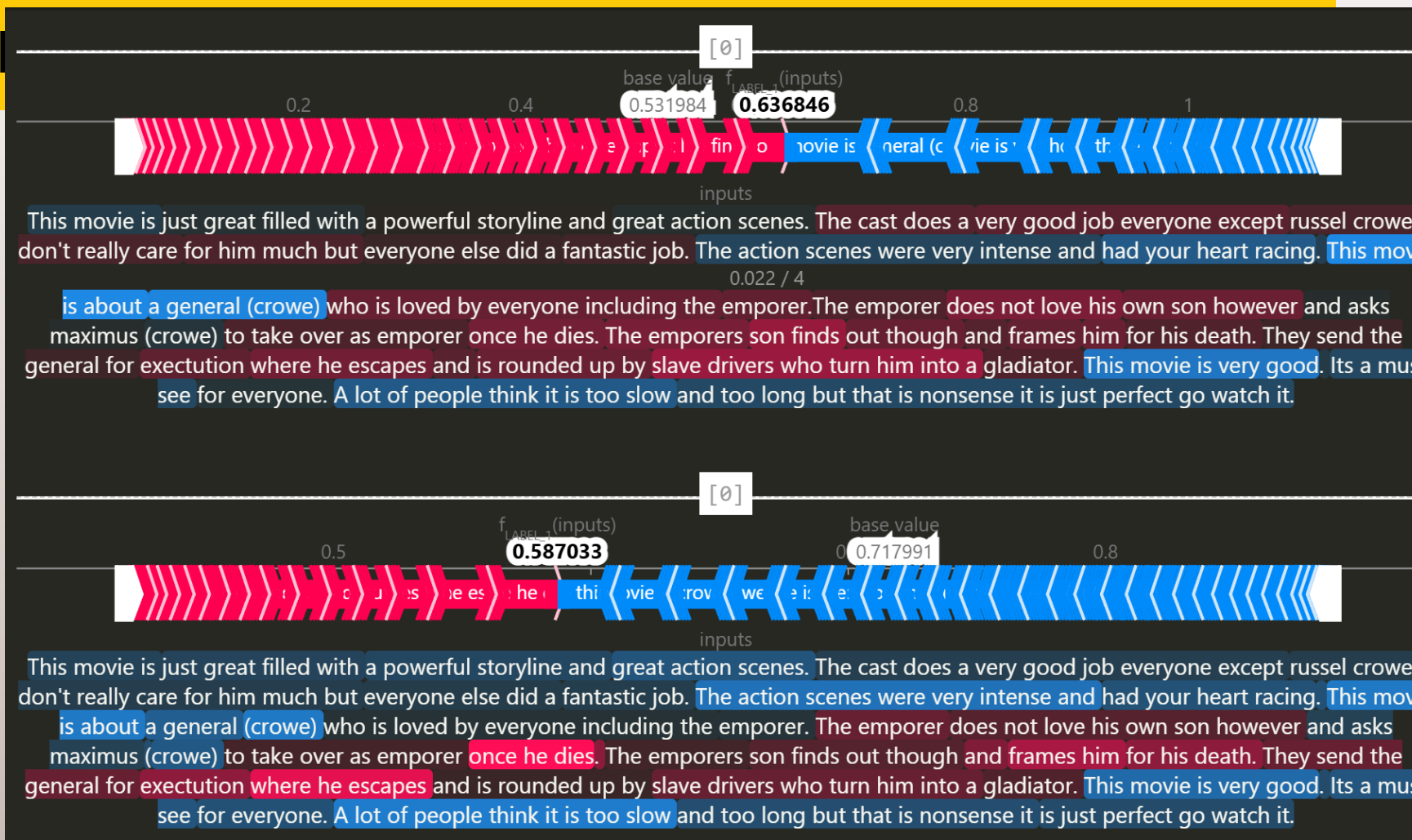
- Random test subset; head-wise attention diagnostics
- Metrics: entropy, span, sequential focus, special-token focus
- Teacher layers 3/6/9/12 vs student layers 1/2/3/4
- Matched pairs show similar heatmaps

Multi-Head Attention Analysis: Teacher (distilled layers) vs Student



RESULTS: ALIGNMENT

- SHAP token attributions on a tricky no-spoiler review
- Both models predict the spoiler class Teacher attribution is more diffuse across the summary
- Student attribution is more local and plot-focused



CONCLUSIONS AND FUTURE WORK

■ Conclusions

- Distillation was largely successful: only **3.74%** PR-AUC drop
- Large efficiency gains: **-86.76%** parameters, **-77.81%** inference time
- Best efficiency–performance trade-off among compared models
- But spoiler detection remains ambiguous and hard (PR-AUC 0.63); false negatives still matter

■ Future Work

- Deploy the model in a browser extension
- Explore better long-review handling beyond head+tail truncation
- Further tune distillation, calibration, and data augmentation
- Use stronger interpretability tools such as Attention Rollout / Attention Flow

See more here: <https://github.com/MarcoColan01/nlp.project-Colangelo-2526>

**THANKS FOR YOUR
ATTENTION!**